

Problem

Find $f(t)$ for each $F(s)$:

(a) $\frac{10s}{(s+1)(s+2)(s+3)}$

(b) $\frac{2s^2 + 4s + 1}{(s+1)(s+2)^3}$

(c) $\frac{s+1}{(s+2)(s^2+2s+5)}$

Step-by-step solution

Step 1 of 11

(a)

Consider the Laplace transfer function of $f(t)$ is,

$$F(s) = \frac{10s}{(s+1)(s+2)(s+3)}$$

The denominator of $F(s)$ has three simple poles.

Use partial fraction expansion to decompose $F(s)$ as follows:

$$\frac{10s}{(s+1)(s+2)(s+3)} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+3} \dots\dots (1)$$

Residue Method to determine coefficients:

Find coefficient A .

$$\begin{aligned} A &= (s+1)F(s)\Big|_{s=-1} \\ &= \frac{10s}{(s+2)(s+3)}\Big|_{s=-1} \\ &= \frac{10(-1)}{(-1+2)(-1+3)} \\ &= -5 \end{aligned}$$

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Find coefficient B .

$$\begin{aligned} B &= (s+2)F(s)\Big|_{s=-2} \\ &= \frac{10s}{(s+1)(s+3)}\Big|_{s=-2} \\ &= \frac{10(-2)}{(-2+1)(-2+3)} \\ &= 20 \end{aligned}$$

Find coefficient C .

$$\begin{aligned} C &= (s+3)F(s)\Big|_{s=-3} \\ &= \frac{10s}{(s+1)(s+2)}\Big|_{s=-3} \\ &= \frac{10(-3)}{(-3+1)(-3+2)} \\ &= -15 \end{aligned}$$

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Substitute -5 for A , 20 for B , -15 for C in equation (1).

$$F(s) = \frac{-5}{s+1} + \frac{20}{s+2} + \frac{-15}{s+3}$$

Apply inverse Laplace Transform on both sides.

$$f(t) = -5e^{-t}u(t) + 20e^{-2t}u(t) - 15e^{-3t}u(t)$$

Therefore, the inverse Laplace transform $f(t)$ of $F(s)$ is $\boxed{(-5e^{-t} + 20e^{-2t} - 15e^{-3t})u(t)}$.

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(b)

Consider the Laplace transfer function of $f(t)$ is,

$$F(s) = \frac{2s^2 + 4s + 1}{(s+1)(s+2)^3}$$

The denominator of $F(s)$ has one simple pole and three repeated simple poles.

Use partial fraction expansion to decompose $F(s)$ as follows:

$$\frac{2s^2 + 4s + 1}{(s+1)(s+2)^3} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{(s+2)^2} + \frac{D}{(s+2)^3} \dots\dots (2)$$

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Residue Method to determine coefficients:

Find coefficient A .

$$\begin{aligned} A &= (s+1)F(s)\Big|_{s=-1} \\ &= \frac{2s^2+4s+1}{(s+2)^3}\Big|_{s=-1} \\ &= \frac{2(-1)^2+4(-1)+1}{(-1+2)^3} \\ &= -1 \end{aligned}$$

Find coefficient D .

$$\begin{aligned} D &= (s+2)^3 F(s)\Big|_{s=-2} \\ &= \frac{2s^2+4s+1}{(s+1)}\Big|_{s=-2} \\ &= \frac{2(-2)^2+4(-2)+1}{(-2+1)} \\ &= -1 \end{aligned}$$

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Find coefficient C .

$$\begin{aligned} C &= \frac{d}{ds} \left[(s+2)^3 F(s) \right] \Big|_{s=-2} \\ &= \frac{d}{ds} \left[\frac{2s^2+4s+1}{(s+1)} \right] \Big|_{s=-2} \\ &= \frac{(s+1)(4s+4) - (2s^2+4s+1)(1)}{(s+1)^2} \Big|_{s=-2} \\ &= \frac{(-2+1)(4(-2)+4) - (2(-2)^2+4(-2)+1)}{(-2+1)^2} \\ &= 3 \end{aligned}$$

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Find coefficient B .

$$\begin{aligned}
 B &= \frac{1}{2!} \frac{d^2}{ds^2} \left[(s+2)^3 F(s) \right] \Big|_{s=-2} \\
 &= \frac{1}{2} \frac{d^2}{ds^2} \left[\frac{2s^2 + 4s + 1}{(s+1)} \right] \Big|_{s=-2} \\
 &= \frac{1}{2} \frac{d}{ds} \left[\frac{(s+1)(4s+4) - (2s^2 + 4s + 1)}{(s+1)^2} \right] \Big|_{s=-2} \\
 &= \frac{1}{2} \frac{d}{ds} \left[\frac{(4s^2 + 8s + 4) - (2s^2 + 4s + 1)}{(s+1)^2} \right] \Big|_{s=-2} \\
 &= \frac{1}{2} \frac{d}{ds} \left[\frac{(2s^2 + 4s + 3)}{(s+1)^2} \right] \Big|_{s=-2} \\
 &= \frac{1}{2} \left[\frac{(s+1)^2(4s+4) - (2s^2 + 4s + 3)2(s+1)}{(s+1)^4} \right] \Big|_{s=-2} \\
 &= \frac{1}{2} \left[\frac{[(-2+1)^2(4(-2)+4)] - [(2(-2)^2 + 4(-2) + 3)2(-2+1)]}{(-2+1)^4} \right] \Big|_{s=-2} \\
 &= 1
 \end{aligned}$$

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Substitute -1 for A , 1 for B , 3 for C , -1 for D in equation (2).

$$F(s) = -\frac{1}{s+1} + \frac{1}{s+2} + \frac{3}{(s+2)^2} - \frac{1}{(s+2)^3}$$

Apply inverse Laplace Transform.

$$f(t) = -e^{-t}u(t) + e^{-2t}u(t) + 3te^{-2t}u(t) - \frac{1}{2}t^2e^{-2t}u(t)$$

Therefore, the inverse Laplace transform $f(t)$ of $F(s)$ is $\boxed{\left(-e^{-t} + \left(1 + 3t - \frac{1}{2}t^2\right)e^{-2t}\right)u(t)}.$

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(c)

Consider the Laplace transfer function of $f(t)$ is,

$$F(s) = \frac{s+1}{(s+2)(s^2+2s+5)}$$

The denominator of $F(s)$ has one simple pole and two complex conjugate poles.

Use partial fraction expansion to decompose $F(s)$ as follows:

$$\frac{s+1}{(s+2)(s^2+2s+5)} = \frac{A}{s+2} + \frac{Bs+C}{s^2+2s+5} \dots\dots (3)$$

Algebraic method to determine coefficients as follows:

$$s+1 = A(s^2+2s+5) + (Bs+C)(s+2)$$

$$\text{if } s = -2,$$

$$5A = -1$$

$$A = \frac{-1}{5}$$

$$\text{if } s = 0,$$

$$5A + 2C = 1$$

$$-1 + 2C = 1$$

$$C = 1,$$

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$$\text{if } s = 1,$$

$$8A + 3B + 3C = 2$$

$$-\frac{8}{5} + 3B + 3 = 2$$

$$3B = -1 + \frac{8}{5}$$

$$3B = \frac{3}{5}$$

$$B = \frac{1}{5}$$

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Substitute $-\frac{1}{5}$ for A , $\frac{1}{5}$ for B , 1 for C in equation (3).

$$F(s) = \frac{-\frac{1}{5}}{s+2} + \frac{\frac{1}{5}s+1}{s^2+2s+5}$$

$$F(s) = \frac{-0.2}{s+2} + \frac{0.2s+1}{(s+1)^2+(2)^2}$$

$$F(s) = \frac{-0.2}{s+2} + 0.2 \left(\frac{s+5}{(s+1)^2+(2)^2} \right)$$

$$F(s) = \frac{-0.2}{s+2} + 0.2 \left(\frac{s+1+4}{(s+1)^2+(2)^2} \right)$$

$$F(s) = \frac{-0.2}{s+2} + 0.2 \left(\frac{s+1}{(s+1)^2+(2)^2} + \frac{4}{(s+1)^2+(2)^2} \right)$$

$$F(s) = \frac{-0.2}{s+2} + 0.2 \left(\frac{s+1}{(s+1)^2+(2)^2} + 2 \left(\frac{2}{(s+1)^2+(2)^2} \right) \right)$$

Apply inverse Laplace transform.

$$f(t) = -0.2e^{-2t}u(t) + 0.2(\cos 2t + 2 \sin 2t)e^{-t}u(t)$$

Therefore, the inverse Laplace transform $f(t)$ of $F(s)$ is,

$$\boxed{\left[-0.2e^{-2t} + 0.2(\cos 2t + 2 \sin 2t)e^{-t} \right] u(t)}.$$